

Equation Sheet

Unit conversions				
$N = \text{kg}\cdot\text{m}/\text{s}^2$	$J = N\cdot\text{m}$	$\text{Pa} = \text{N}/\text{m}^2 = \text{J}/\text{m}^3$	$1 \text{ L} = 1\cdot 10^{-3} \text{ m}^3$	$1 \text{ ml} = 1 \text{ cm}^3$
Kelvin = Celsius+273	Fahrenheit = (Celsius*9/5)+32	Area circle = πr^2	Volume sphere = $(4/3)\pi r^3$	Surface area sphere = $4\pi r^2$
Constants				
Room temp = 23°C	1Cal = 1000 cal = 4.18 kJ	Gravitational acceleration ($g = -9.8 \text{ m/s}^2$)	Avogadro's number ($6.022\cdot 10^{23}$ molecules/mol)	
Boltzmann constant $k_B = (1.38\cdot 10^{-23} \text{ J/K})$	Specific heat water 4.19 J/gK	Latent heat of vaporization for water 2260 J/g	Latent heat of fusion for water 334 J/g	
1 mol = 6.022×10^{23} molecules	gas constant = 22.4L/mol	D_{heat} in water $1.4\cdot 10^{-3} \text{ cm}^2/\text{sec}$	D_{Ca} $8.0\cdot 10^{-10} \text{ m}^2/\text{sec}$	

Equation	Description		Units
$L = \sqrt{4Dt}$	L	Distance of diffusion	m
	D	Diffusion coefficient	m^2/s
	t	Time to diffuse	s
$\phi = \frac{V_{\text{dispersed}}}{V_{\text{total}}}$	ϕ	Volume fraction	-
	$V_{\text{dispersed}}$	Volume of dispersed phase	L
	V_{total}	Total volume	L
$E = \frac{\sigma}{R}(\phi - \phi_c) \quad \phi > \phi_c$	E	Elastic modulus	Pa
	σ	Surface tension	N/m
	R	Droplet radius	m
	ϕ	Dispersed phase volume fraction	-
	ϕ_c	Critical volume fraction (0.64 for spheres)	-

Equation		Description	Units
$\text{pH} = -\log_{10}[\text{H}^+]$	$[\text{H}^+]$	Concentration of protons (also known as H^+ ions) in solution	mol/L
	pH	A measure of acidity	none
$Q = mc_p \Delta T$	Q	Heat	J
	m	Mass	g
	c_p	Specific heat capacity	J/g K or J/g°C
	ΔT	Change in temperature	K or °C
$Q = mL_v$ $Q = mL_f$	Q	Heat	J
	m	Mass	g
	L_v	Latent heat of vaporization	J/g
	L_f	Latent heat of fusion	J/g
$U = Ck_B T$	U	Interaction energy for liquid to gas transitions	J
	C	Constant (1.5 for water)	none
	k_B	Boltzmann constant	J/K
	T	Temperature	K
$E = \frac{F / A}{\Delta L / L_0}$	E	Elastic modulus	Pa
	F	Force	N
	A	Area over which force is applied	m ²
	ΔL	Change in length	m
	L_0	Initial length	m
$E = \frac{U_{\text{int}}}{l^3} = \frac{k_B T}{l^3}$	E	Elastic modulus	Pa
	k_B	Boltzmann constant	J/K
	T	Temperature	K
	l	Average distance between crosslinks	m
$N(t) = N_0 e^{kt}$ where $k = \frac{\ln 2}{\tau}$	$N(t)$	Number of cells at time t	microbes
	N_0	Number of cells at time 0	microbes
	k	Growth or death rate of cells	s ⁻¹
	t	Time elapsed	s
	τ	Doubling time	s